MAQAM AI | Technical Evidence Brief

Independent reproducibility snapshot • 11 November 2026 • Not an internship motivation statement

Project: AI-assisted research software for formal score processing and computational musicology. The released code includes a Next.js interface, Python/FastAPI API and exact dynamic-programming components. It is an engineering and formal-reasoning prototype, not a validated system for singing, pronunciation or music-alignment accuracy.

Verified now

Check	Outcome
Focused FastAPI / score ingestion / identifiability tests	35/35 passed (11 Nov 2026)
Original release full-test evidence	376/376 passed (4 Nov 2026; historical)
Constructed score, tie continuation	3 written chords → 2 normalized events
Formal chord positions on constructed sample	[0, 2]
Identical output when score lyrics removed	Yes; hidden lyric text not returned
DTD-containing request	Rejected with HTTP 422
Real sung-audio / pronunciation accuracy	Not evaluated

One-command reproducibility

Run **python portfolio_smoke_demo.py** from the source root. It exercises the existing in-process FastAPI endpoint without starting a web server. Run the four listed pytest modules in **START_HERE_PORTFOLIO.md** for the 35 focused tests. The Next.js UI is present as source; a current production build or deployed URL was not verified.

Research integrity and limitations

The exact DP solves its formal cost. On genuine public-score-derived tests, many formal alignments are tied or inconsistent with lyric annotations. A notation-aware heuristic matched two eligible songs but did not meet the prespecified coverage criterion. Original scores were inspected through authenticated GitHub access; unchanged full real XML was not executed by the local Python adapter. These limitations are part of the project evidence, not defects to conceal.

CaCTüS | Submission-readiness gate

Official organiser: Max Planck Institutes and Tübingen AI Center | Applications close 16 November 2026.

First: academic eligibility

The internship is for eligible Bachelor/Master students or certain graduates. Applicants who have not completed a Bachelor must have finished at least four university semesters before the July 2027 internship. The internship requires full on-site availability 1 July to 30 September 2027. If you are still in school, confirm eligibility before investing days in a submission.

Second: the official application rules

Applicants choose from the host's advertised project options; MAQAM can be mentioned as a previous project, not proposed as a replacement host project. Required materials include an academic CV of no more than two pages, university transcripts, applicant-authored motivation responses, and optional reference letter. Only complete applications submitted through the portal count; look for the automatic confirmation email.

Important authorship boundary

CaCTüS explicitly permits AI assistance for clarity, grammar and style, but not for inventing ideas, concepts, original intellectual content, or personal experiences. This is an AI-assisted technical evidence brief, not a self-authored application statement. The applicant must verify their individual role and be able to explain each claimed skill or outcome.

Three working days: responsible execution

• Day 1: Confirm academic eligibility; obtain the up-to-date host projects and university transcripts; identify true individual MAQAM contributions.

• Day 2: Prepare a truthful CV and applicant-written motivation; use language editing only. Independently run the project smoke demonstration.

• Day 3: Review all materials against the official portal; complete and submit before 16 November; verify the automated submission confirmation.

Official programme: https://projects.tuebingen.mpg.de/program

Official FAQ: https://projects.tuebingen.mpg.de/faq

Registration: https://cactus-apply.tue.mpg.de/registration/cactus-2027

Source evidence: local MAQAM V10 Timing Integrity integrated release (4 November 2026), with existing model, API, tests and published limitation records.